

Lec 51

Concentration Inequalities.

① Markov Inequality

For a non-negative random variable X ,

$$P(X > a) \leq \frac{E[X]}{a} \quad \text{for } a > 0.$$

$$\Leftrightarrow a \cdot P(X > a) \leq E[X].$$

X is a discrete R.V

$$E[X] = \sum_x x P_X(x)$$

$$= \sum_{x > a} x P_X(x) + \underbrace{\left(\sum_{a \leq x \leq a} x P_X(x) \right)}_{\geq 0}$$

$$\geq \sum_{x > a} x P_X(x) > a \sum_{x > a} P_X(x) = a P(X > a).$$

Total law of expectation

$$E[X] = E[X|A]P(A) + E[X|A^c]P(A^c)$$

$$A = \{X > a\}, \quad A^c = \{X \leq a\}$$

$$E[X|X > a] > a$$

$$a \geq E[X|X \leq a] \geq 0$$

$$\Rightarrow E[X] > a P(A) + 0 \times P(A^c)$$

$$\Rightarrow P(A) = P(X > a) < \frac{E[X]}{a}$$

$$f_{X|A}(x) = \begin{cases} \frac{f_X(x)}{P(A)} & x > a \\ 0 & \text{otherwise} \end{cases}$$

$$\Rightarrow E[X|A] > a$$

$$E[X|A] = \int_a^\infty x f_{X|A}(x) dx$$

$$= \int_0^\infty x f_{X|A}(x) dx$$

$$\geq a \int_0^\infty f_{X|A}(x) dx = a$$

$x < 0$ & $x > 0$
 $\int_{-\infty}^{\infty} f_X(x) dx = 1$



≥ 0
 (x)

$a P(X > a)$

$x \in \mathbb{R}$ st $0 \leq x \leq b$
 then $E[X] \leq b$



$$P(X > a) = 1 - F_X(a)$$

$$\int_{-\infty}^{\infty} x f_X(x) dx$$

$$\int_{-\infty}^{\infty} x f_X(x) dx$$

$$\int_{-\infty}^{\infty} x f_X(x) dx = a$$

Examples

(a) $X \sim \text{Binomial}(n, p)$

$$P(X \geq n) \leq E[X]$$

 $E[X]$

 Area $(1 - F_X(a))a =$

$$E[X] = \int_0^{\infty} (1 - F_X(x)) dx$$

When is Markov inequality met with equality:

$$P(X \geq a) \leq E[X]/a$$

Examples

$$(a) X \sim \text{Binomial}(n, p) \quad E[X] = np \quad 1 > p > 0$$

$$P(X \geq an) \leq \frac{E[X]}{an} = \frac{np}{an} = \frac{p}{a}$$

$$\text{Suppose } p = 1/2, a = 3/4$$

$$P(X \geq \frac{3}{4}n) \leq \frac{1/2}{3/4} = \frac{2}{3}$$

$$\text{Area } (1 - F_X(a))a = P(X \geq a)a$$

$$E[X] = \int_0^{\infty} (1 - F_X(x)) dx$$

When is Markov inequality met with equality: (Exercise)

$$P(X \geq a) \leq \frac{E[X]}{a}$$

$$(b) X \sim \text{Uniform}[0, 4] \quad E[X] = 2$$

$$P(X \geq 2) \leq \frac{2}{2} = 1$$

$$P(X \geq 3) \leq \frac{2}{3}$$

$$P(X \geq 4) \leq \frac{2}{4} = \frac{1}{2}$$

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Concentration Inequalities.

② Chebyshev Inequality

X is a random variable with finite Variance
and finite mean μ .

$$P(|X - \mu| > \epsilon) \leq \frac{\text{Var}(X)}{\epsilon^2}$$

$$\begin{aligned} \Rightarrow X - \mu > \epsilon &\Rightarrow X > \mu + \epsilon \\ -(X - \mu) > \epsilon &\Rightarrow X < \mu - \epsilon \end{aligned}$$

$$P(X \notin [\mu - \epsilon, \mu + \epsilon]) \leq \frac{\text{Var}(X)}{\epsilon^2}$$

$$Y = (X - \mu)^2$$

$$P(|X - \mu| > \epsilon) = P(Y > \epsilon^2) \leq \frac{E[Y]}{\epsilon^2} = \frac{\text{Var}(X)}{\epsilon^2}$$

Markov inequality
↓

Example

$$P(X > \mu)$$

$$A = \{ \omega \in \Omega : \dots \}$$

$$B = \{ \omega \in \Omega : \dots \}$$

$\omega \in A$

A

Examples

$$\omega = \{0, 1, \dots, n\}$$

$$X \sim \text{Binomial}(n, p) \quad 0 < p < 1$$

$$\begin{aligned} P(X > kn) &= P(X - np > (k-p)n) \\ &\leq P(|X - np| > (k-p)n) \leq \frac{\text{Var}(X)}{(k-p)^2 n^2} \end{aligned}$$

Chaplyshin

$$A = \{\omega \in \omega \mid X(\omega) - np > (k-p)n\}$$

$$B = \{\omega \in \omega \mid |X(\omega) - np| > (k-p)n\}$$

$$\omega \in A \Rightarrow \omega \in B$$

$$A \subseteq B$$

$$\leq \frac{n \text{Var}(X)}{n^2 (k-p)^2}$$

$$= \frac{p(1-p)}{n(k-p)^2}$$

$$k = 3/4, p = 1/2$$

$$P(X)$$

$$P(X)$$

$$1 - p = n - 1$$

$$x < 1$$

$$\leq \frac{\text{Var}(X)}{(x-p)^2}$$

Chebyshev

$$\leq \frac{\text{Var}(X_1)}{n(x-p)^2}$$

$$= \frac{p(1-p)}{n(x-p)^2}$$

$$x = 3/4, p = 1/2$$

$$X = X_1 + \dots + X_n$$

X_i 's are iid Bernoulli(p)

$$\begin{aligned} \text{Var}(X) &= \sum_{i=1}^n \text{Var}(X_i) \\ &= n \text{Var}(X_1) \end{aligned}$$

$$P(X > \alpha n) \leq \frac{1/4}{n(1/4)^2} = \frac{4}{n}$$

Chebyshev inequality

$$P(X > \alpha n) \leq 2/3$$

Markov inequality

(b)

✓
Suppose X is bounded in $[0, c]$

$$\text{Var}(X) \leq \boxed{\frac{c^2}{4}}$$

$$P(|X - \mu| > \varepsilon) \leq \frac{c^2}{4\varepsilon^2}$$

$$\begin{aligned}\text{Var}(X) &= E[X^2] - (E[X])^2 \\ &\leq E[cX] - (E[X])^2 \\ &= E[X][c - E[X]] \\ &\leq \left(\frac{E[X] + c \cdot E[X]}{2} \right)^2\end{aligned}$$

Suppose Y is bounded in $[a, b]$

$$ab \leq \left(\frac{a+b}{2} \right)^2$$

$$\Rightarrow P(|Y - E[Y]| > \varepsilon) \leq \frac{(b-a)^2}{4\varepsilon^2}$$

$$X = Y - a \in [0, b-a]$$

$$\begin{aligned}P(|Y - E[Y]| > \varepsilon) &= P(|X + a - E[X]| > \varepsilon) \\ &\leq \frac{(b-a)^2}{4\varepsilon^2}\end{aligned}$$

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Concentration Inequalities:

③ Weak Law of Large numbers.

$X_1, X_2, \dots, X_n$ that are iid
 $E[X_i], \text{Var}(X_i)$ are finite.

$$S_n = \underbrace{X_1 + X_2 + \dots + X_n}_n$$

$$E[S_n] = \frac{n E[X_1]}{n} = E[X_1]$$

$$\text{Var}[S_n] = \text{Var}\left(\underbrace{X_1 + \dots + X_n}_n\right)$$

$$\begin{aligned} &= \frac{1}{n^2} \text{Var}(X_1 + X_2 + \dots + X_n) \\ &= \frac{1}{n^2} \times n \cdot \text{Var}(X_1) = \frac{\text{Var}(X_1)}{n} \end{aligned}$$

l.i.d.
nature of
 $X_1, \dots, X_n$.

$$P(|S_n - E[S_n]| > \epsilon) \stackrel{\text{Chebyshev}}{\leq} \frac{\text{Var}(S_n)}{\epsilon^2}$$

$$= \frac{\text{Var}(X_1)}{n\epsilon^2}$$

For any $\epsilon > 0$,

$$0 \leq \lim_{n \rightarrow \infty} P(|S_n - E[X_i]| > \epsilon) \leq \lim_{n \rightarrow \infty} \frac{\text{Var}(X_1)}{n\epsilon^2} = 0$$

$$\begin{array}{ccccccc} a_1 & a_2 & a_3 & \dots & a_n & \dots & \\ b_1 & b_2 & b_3 & \dots & b_n & \dots & \end{array}$$

$$a_n \leq b_n \quad \forall n$$

$$\lim_{n \rightarrow \infty} a_n \leq \lim_{n \rightarrow \infty} b_n$$

$$\lim_{n \rightarrow \infty} P(|S_n - E[X_i]| > \epsilon) = 0$$

Suppose you toss a coin with bias p , n # of times.

You want to be 99.9% sure that the sample estimate you have is within an error of $\epsilon = 10^{-3}$. How many samples do you need to ensure this?

Prob of failure should be within $\frac{0.01}{100} = 8 \times 10^{-4}$

find n st

$$P(|S_n - p| > \epsilon) \leq \delta$$

We know $P(|S_n - p| > \epsilon) \leq \frac{p(1-p)}{n\epsilon^2}$

If $\frac{p(1-p)}{n\epsilon^2} < \delta$ - We are guaranteed

Since $p \leq \frac{1}{2}$, n # of times.

sure that the

within an error of

you need to ensure
this?

and be within $\frac{0.01}{100} = 8$
" 10^4 .

$$\leq 8$$

$$\leq \frac{p(1-p)}{n\epsilon^2}$$

We are guaranteed

$$n > \frac{p(1-p)}{\epsilon^2 \delta}$$

$$= \frac{1}{4} \cdot \frac{1}{10^{-6} \times 10^{-4}}$$

$$= \frac{10^{10}}{4}$$